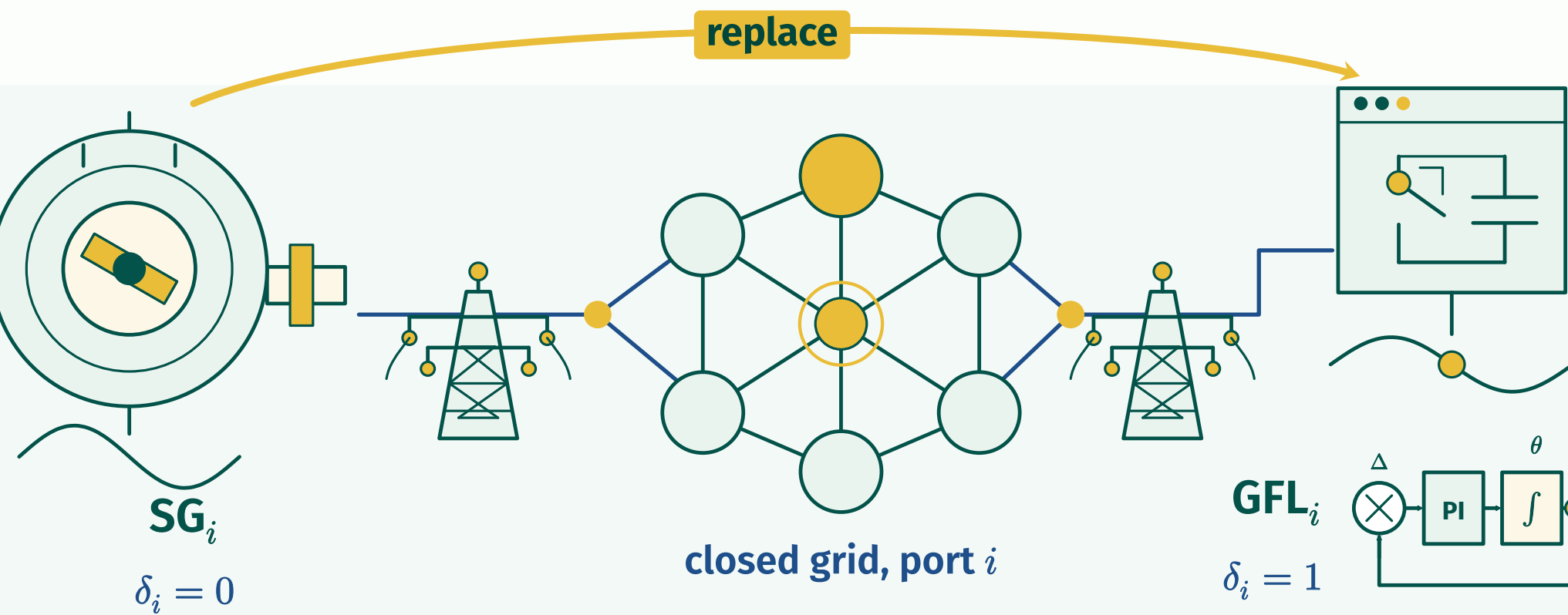


1 THE PROBLEM: SAFE SG → GFL TRANSITION

A replacement is local hardware, but stability is a property of the closed grid and its controller architecture.



A safe SG → GFL transition is a portfolio-controller co-design problem.

Engineering objective

$$\max_{S, K} \sum_{i \in S} P_i^0 \quad \text{s.t.} \quad \alpha_{\perp}(S, K) \leq -\sigma_{\text{req}}, \quad K \in \mathcal{K}_n$$

P_i^0 SG, dispatch $\alpha_{\perp}$ rightmost non-gauge root σ_{req} decay margin $\mathcal{K}_n$ gains using $\leq n$ hops

- Choose portfolio δ
- Choose radius $n: K \in \mathcal{K}_n$
- Design gains K_p, K_I
- Verify grid spectrum, events

Report the largest independently verified portfolio found.
Certified global optimality is future work.

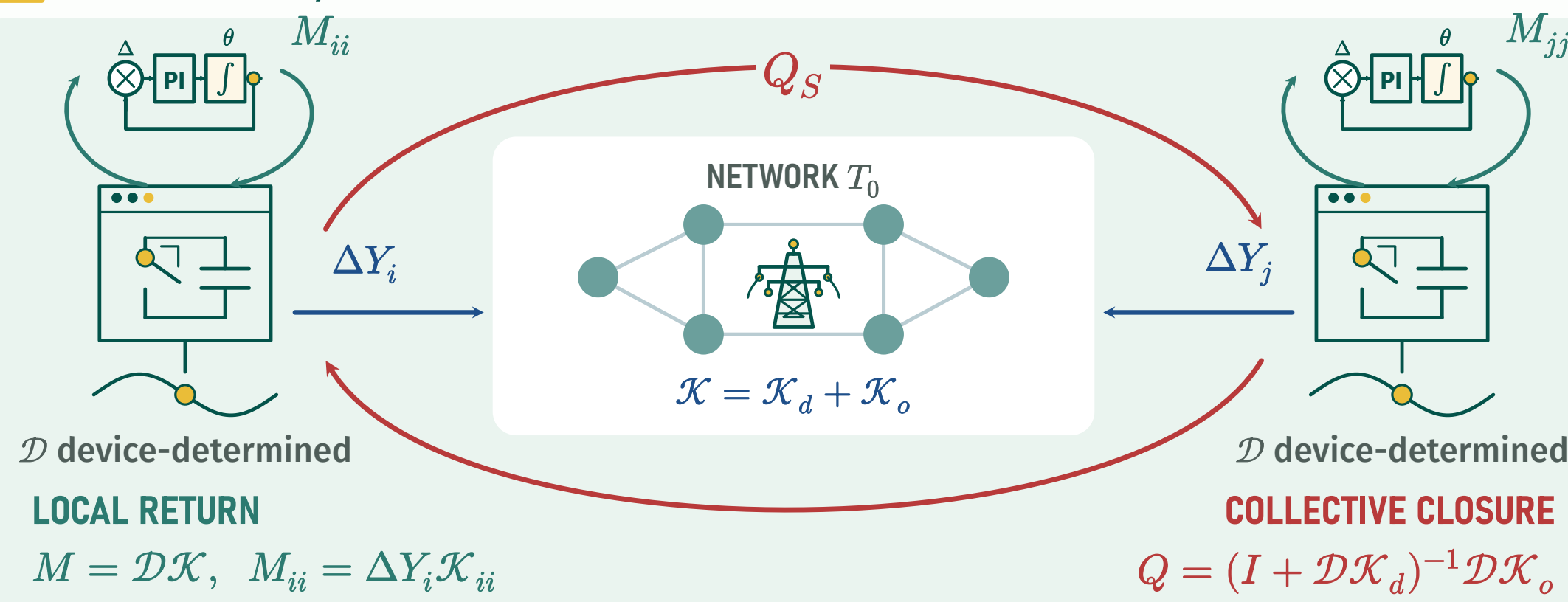
→ Model it

4 EXACT NETWORK CLOSURE: LOCAL × COLLECTIVE

$$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$$

$$\Delta Y_i = E_i^T [T_{\{i\}}(s) - T_0(s)] E_i, \quad \mathcal{K} = E^T T_0^{-1} E = \mathcal{K}_d + \mathcal{K}_o$$

A Two devices, one network



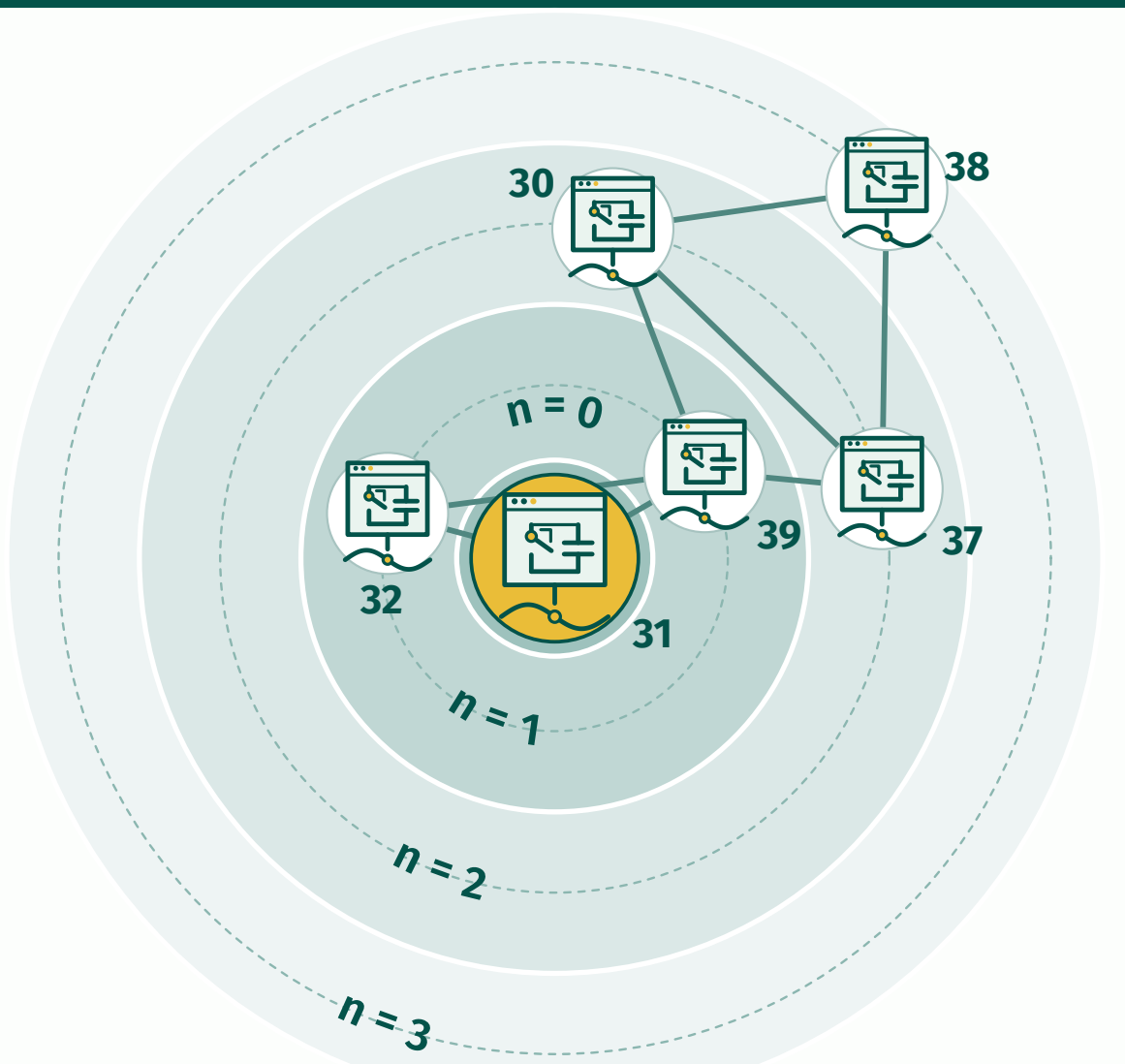
$$\det(I + M_S) = \prod_{i \in S} \det(I + M_{ii}) \cdot \det(I + Q_S) \quad \text{EXACT}$$

Audit: stored boundary $H_4 = \{30, 33, 35, 37\}$, $f \approx 0.7064$ Hz
 $\min_i \sigma_{\min}(I + M_{ii}) = 0.4892$, $\sigma_{\min}(I + Q_{H_4}) = 4.39 \times 10^{-8}$
Delayed full IEEE-39, 44 ms: $\text{eig}(Q) = -1$ at every unstable root, all $|L_i| \geq 0.011$.

Nothing failed locally. The network closed the destabilizing loop.

→ Who can repair it?

7 INFORMATION RADIUS AND N-HOP CONTROL



$$\mathcal{N}_i^{(n)} = \{j : \text{dist}_{G_c}(i, j) \leq n\}$$

$$C_i^{(n)} k_i = b_i, \quad \epsilon_i(n) = \|(I - C_i C_i^+) b_i\| / \|b_i\|$$

$$\mathcal{K}_n \subset \mathcal{K}_{n+1} \Rightarrow \epsilon(n+1) \leq \epsilon(n)$$

EXACT

free n -hop class only

$$\dot{\xi}_i = \sum_{j \in \mathcal{N}_i^{(n)}} k_{i,j} e_j(t - \tau_j)$$

$$t_f \dot{\omega}_i = \xi_i - \omega_i + \sum_{j \in \mathcal{N}_i^{(n)}} k_{p,i,j} e_j(t - \tau_j)$$

- local $n=0$
- polynomial in L
- free n -hop
- dense benchmark

MAIN CONTRIBUTIONS

- Minimal replacement blockers: the H_4 witness is unstable although all its proper subsets are stable.
- Exact closure factorization of the network return into a local and a collective part.
- Beyond nodal damping: self-energy and signed damping redistribution.

- Grid-embedded PLL gain synthesis and exact delay transport.
- Information radius on the full IEEE-39: authority grows, safety must still be verified.

2 MODEL: DAE → TRANSVERSE STABILITY

$$E(x, \delta) \dot{x} = f(x, z; \delta, K, \tau)$$

$$0 = g(x, z; \delta)$$

x states, z network, δ portfolio, K gains, τ delays
equilibrium: $f(x^*, z^*) = 0, g(x^*, z^*) = 0$

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} f_x & f_z \\ g_x & g_z \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \end{bmatrix}$$

$$A = f_x - f_z g_z^{-1} g_x$$

Schur complement, g_z invertible

$$A_{\perp} = Z^T A Z$$

Z : orthonormal basis of $\gamma^{\perp}$ (gauge $A\gamma = 0$)

$$\alpha_{\perp}(S, K) = \max \text{Re } \sigma(A_{\perp})$$

With PLL delay, $\alpha_{\perp}$ is the rightmost root of $\det \Delta_{\perp}(s)$;
 $\Delta(s) = sI - A_0 - B E(s) C$, exact exponentials.

< 0 STABLE

> 0 UNSTABLE



→ Where it fails

5 BEYOND NODAL DAMPING: EFFECTIVE OPERATORS

A Exact Schur self-energy

$$S_r(s) = sI - A_{rr} - A_{rc}(sI - A_{cc})^{-1} A_{cr}$$

$$T(s) = s^2 M + s D_0 + L + \Pi(s)$$

r kept, c eliminated

B Even/odd split $\Pi = \Pi_e + \Pi_o$

$$L_{\text{eff}} = L + \Pi_e(0), \quad D_{\text{eff}} = D_0 + \Pi_o(s)/s$$

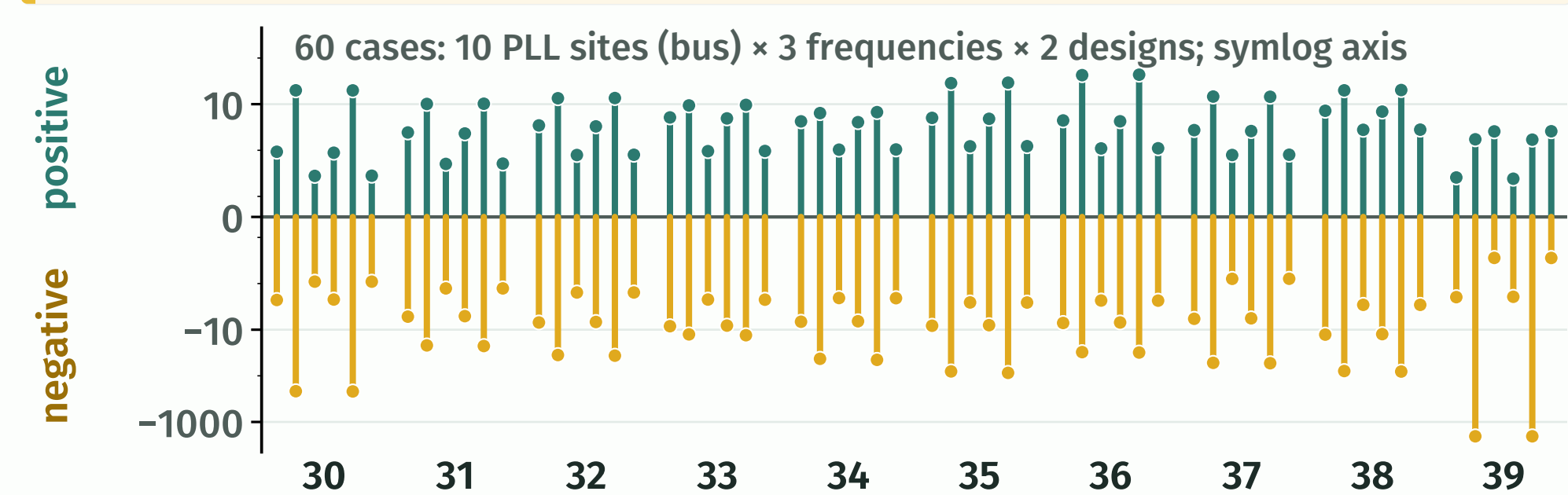
$$M_{\text{eff}} = M + [\Pi_e(s) - \Pi_e(0)]/s^2$$

$$T = s^2 M_{\text{eff}} + s D_{\text{eff}} + L_{\text{eff}}$$

M_{eff} : an operator, not rotor inertia.

C Damping operator $D_H(\Omega) = \text{Herm } Z(j\Omega)$

$$\delta D_H = \frac{1}{2}(a v^H + v a^H) \quad \text{one positive, one negative eigenvalue}$$



A local change moves damping between paths.

damping ↑
damping ↓

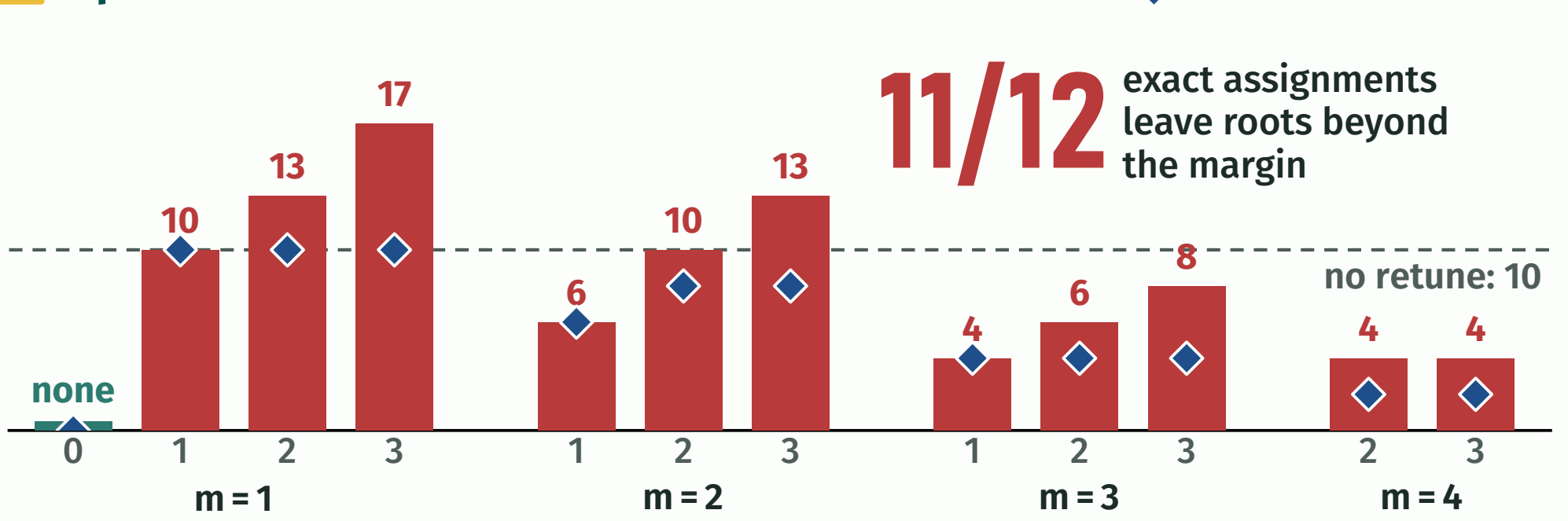
→ Damping alone misleads

8 N-HOP RESULTS: AUTHORITY, SPILLOVER, REPAIR

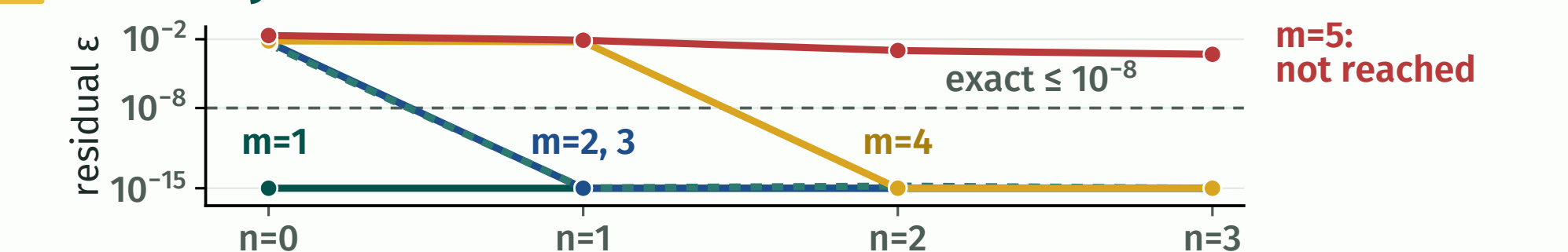
Full IEEE-39, exact delay 44 ms

NUMERICAL OBSERVATION

B Spillover on other roots



A Authority: residual vs radius



Each extra mode pinned needs more measured neighbors ($2m \leq 2|\mathcal{N}_i|$).

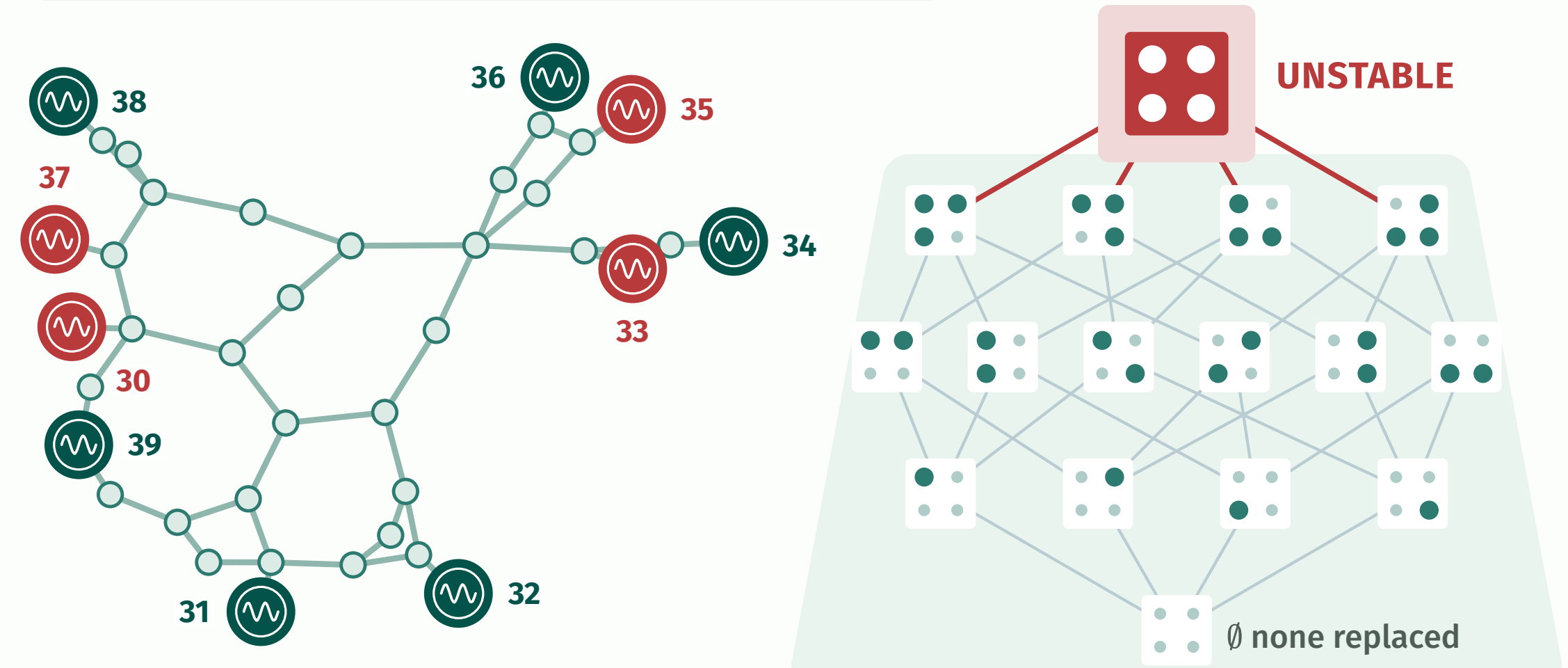
Repair at 44–52 ms already works with $n=0$. Largest verified share: 90 % for every n ; 92.5 % fails for all n . Declared sweep, not a certified optimum.

More modal authority $\neq$ a safer grid.

→ Does it hold nonlinearly?

3 IEEE-39: MINIMAL COLLECTIVE BLOCKER

SEPARATE TX4 FIXED-POLICY MODEL CONTRACT (NO PLL DELAY)



IEEE-39: red = the four sites of H_4 , green = other SGs

glyph = one portfolio; filled dot = replaced site
dot positions: 30 33 (top), 35 37 (bottom)

Minimal collective blocker $H_4 = \{30, 33, 35, 37\}$

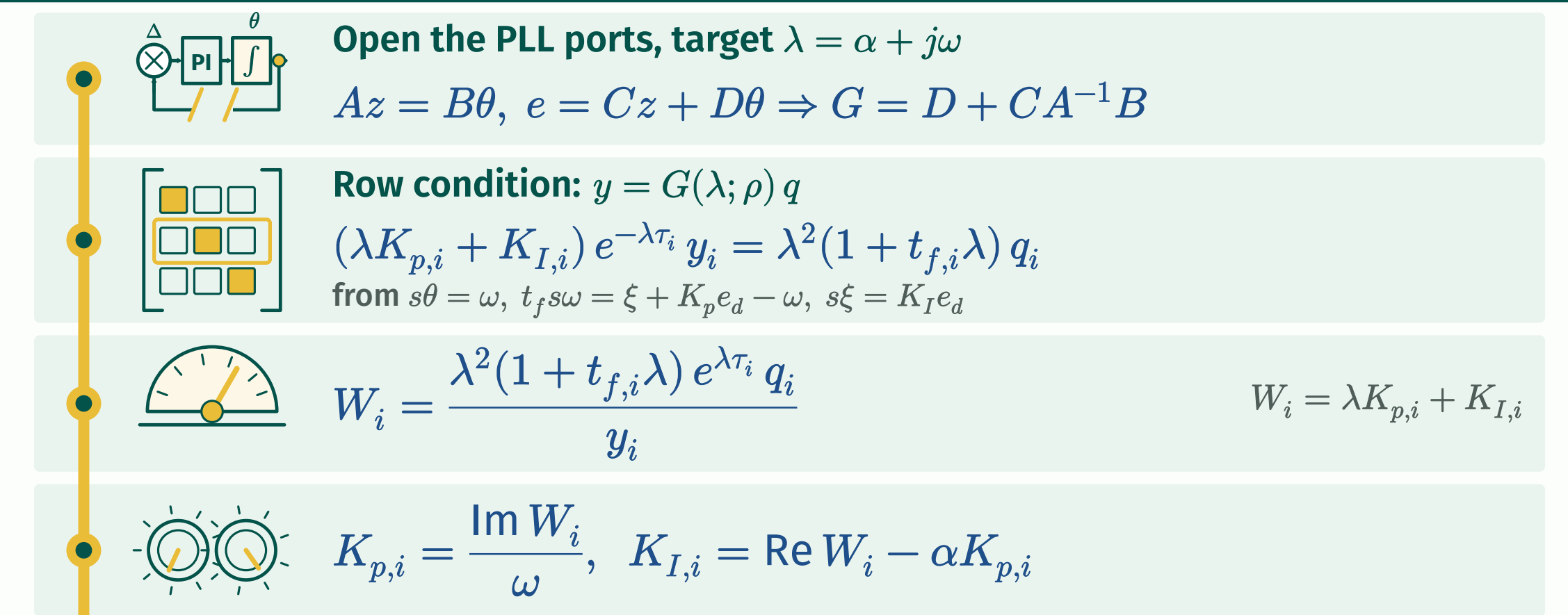
$$\alpha_{\perp}(R) < 0 \quad \forall R \subsetneq H_4, \quad \text{but} \quad \alpha_{\perp}(H_4) = +0.127 \text{ s}^{-1} > 0$$

Every proper subset stays stable (15 of 15); adding the fourth site pushes one mode at $f = 0.6223$ Hz into the right half-plane.

Canonical nominal witness, not a universal blocker.

→ Why? closure

6 GRID-EMBEDDED PLL GAIN SYNTHESIS



$$K = \Phi(\rho, \lambda, q, \tau)$$

The gain is local; its design is embedded in the grid mode.

$$[\lambda K_p(h) + K_I(h)] e^{-\lambda(\tau+h)} = [\lambda K_p + K_I] e^{-\lambda \tau}$$

$$\left. \frac{d\mu}{dh} \right|_0 = -K_p(\mu - \lambda)(\mu - \bar{\lambda}) \frac{\partial \mu}{\partial K_I}$$

Preserving one mode does not preserve the others.

Full IEEE-39, exact delay 44 ms

no retune
8 roots
events abort

protect 4.91 Hz
0 roots
2/5 events pass

protect 1.04 Hz
0 roots
5/5 events pass

abort = integrator stopped

→ With more information

9 DELAYS, CO-DESIGN, AND NONLINEAR CHECKS

A Unequal latencies mix graph modes

$$E_{\tau}(s) = \text{diag } e^{-s\tau_i}, \quad [L, K(L)E_{\tau}] = K(L) [L, E_{\tau}]$$

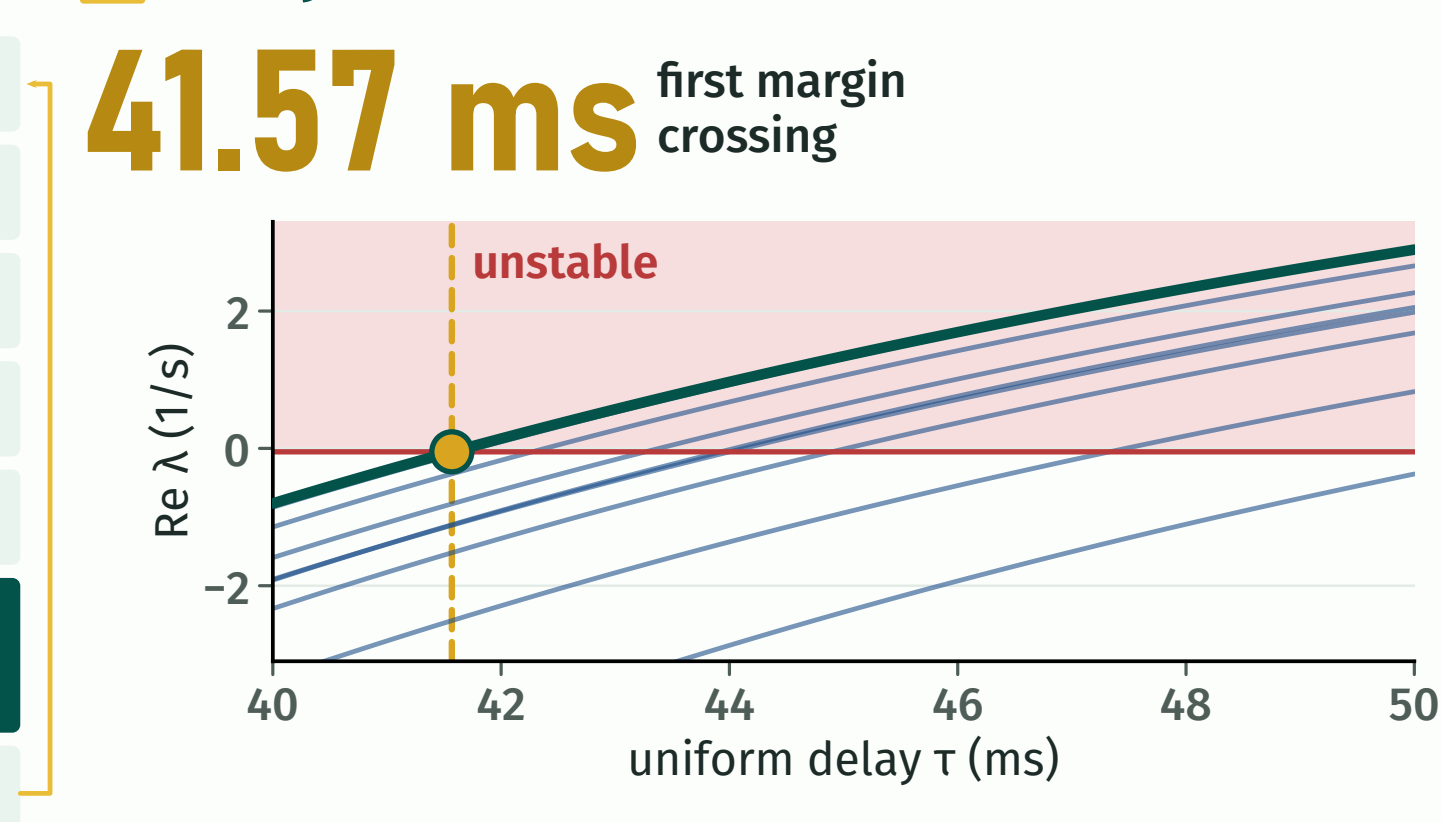
$$\| [L, E_{\tau}(j\Omega)] \|_F^2 = 8 \sum_{i,j \in E} \omega_{ij}^2 \sin^2 \left(\frac{\Omega(\tau_i - \tau_j)}{2} \right)$$

zero only if $\tau_i = \tau_j$ on every edge

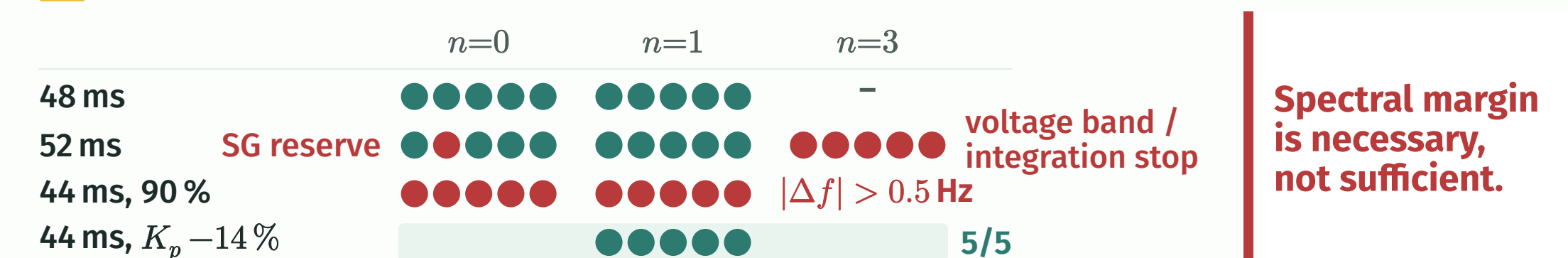
C Co-design loop

- Choose portfolio
- Equilibrium, spectrum
- Closure Q, D_H
- Radius n
- Gain map, predictor
- FULL-SPECTRUM corrector
- Nonlinear events

B Delay root locus



D Five frozen nonlinear events



SCOPE OF THE RESULTS

- EXACT: closure factorization, gain map W_i , delay transport, commutator identity, signed δD_H
- NUMERICAL: H_4 witness (TX4 contract), full IEEE-39 delay campaign, n -hop authority, spillover, repair, nonlinear events
- PROPOSED: $M_{\text{eff}}, D_{\text{eff}}$ graph diagnostics, certified global maximum share, EMT and current limits

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PRESERVING ONE MODE DOES NOT PROTECT THE GRID.

More information gives more control authority — only the full grid decides whether a larger SG → GFL transition is safe.